

The Riemann Hypothesis

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Abstract

Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ denote the Hardy function, and $\Theta(t) = \vartheta(t) + ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. The inverse-phase pullback

$$Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$$

extends to an entire function $\tilde{Y}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$. We prove $\tilde{Y}$ belongs to the Laguerre–Pólya class, hence all zeros of $\tilde{Y}$ are real, hence all real zeros of Z exhaust the zero set of Z under analytic continuation, hence all nontrivial zeros of ζ have real part $\frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy). $Z : \mathbb{R} \rightarrow \mathbb{R}$, $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, $\vartheta(t) = \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Zeros: $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. Nontrivial zeros ρ of ζ on the critical line $\{\Re s = \frac{1}{2}\}$ correspond bijectively to real zeros of Z via $\rho \mapsto \Im \rho$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.

Axiom 3 (Riemann–Siegel). For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$, $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$, $R(t) = O(t^{-1/4})$.

Axiom 4 (Schwartz–Paley–Wiener). Let $u \in \mathcal{S}'(\mathbb{R})$ be a tempered distribution of polynomial order with $\text{supp } \hat{u} \subseteq [-\sigma, \sigma]$. Then u agrees a.e. with the restriction to $\mathbb{R}$ of a unique entire function of exponential type $\leq \sigma$ of polynomial growth on $\mathbb{R}$. Conversely, any entire function of exponential type $\leq \sigma$ of polynomial growth on $\mathbb{R}$ has Fourier transform (as a tempered distribution) supported in $[-\sigma, \sigma]$. (Hörmander, *The Analysis of Linear Partial Differential Operators I*, Theorem 7.3.1.)

Axiom 5 (Band-limit of Y , established in [1]). Let Y be as in Definition 9, extended by 0 to $u < 0$, regarded as a tempered distribution on $\mathbb{R}$. Then

$$\text{supp } \hat{Y} \subseteq [-1, 1].$$

This is the conclusion of [1, Corollary 18], obtained from the structural phase identity $\Phi'_{n,\sigma,\mu}(t) = \Theta'(t)(\sigma\omega_n(t) - \mu)$ together with $\sigma\omega_n(t) \in [-1, 1]$: for every μ with $|\mu| > 1$ the phase $\Phi_{n,\sigma,\mu}$ has no stationary point in t at any mode n or sign σ , hence the Riemann–Siegel mode integrals admit exact integration-by-parts with a denominator $|\Phi'| \geq (|\mu| - 1)\Theta'(t)$ uniformly bounded away from zero, and the Gabcke remainder is controlled uniformly in μ . The weak- $*$ concentration argument of [1, Corollary 18] (using iterated integration by parts in u with cutoff $M = T^{1/4}$, $k = 4$) upgrades the pointwise off-band decay to exact distributional support in $[-1, 1]$.

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Axiom 6 (Reality-of-zeros for sharp-band entire functions). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$ and with $1 \in \text{supp } \hat{F}$ (sharp endpoint), then every zero of F lies in $\mathbb{R}$.

Axiom 7 (Sharp endpoint for Z). $1 \in \text{supp } \hat{Y}$.

2 Construction

Definition 8. $\Theta(t) := \vartheta(t) + ct$ for fixed $c > c_*$.

Definition 9. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, principal positive branch.

Lemma 10 (Warp). $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_* > 0$, and for every $t \in \mathbb{R}$, $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$.

Proof. $\Theta' = \vartheta' + c \geq c - c_* > 0$ by Axiom 2, so Θ is a strictly increasing C^∞ bijection. The identity is Definition 9 solved for Z . $\square$

Lemma 11 (Frequency ratio). $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1]$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n \iff t \geq 2\pi n^2$ by Axiom 2; then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$, and $\vartheta' \rightarrow \infty$. $\square$

3 Entire Extension

Lemma 12 (Polynomial growth of Y). $|Y(u)| = O(|u|^{1/4})$ as $|u| \rightarrow \infty$; in particular Y is a tempered distribution of polynomial order on $\mathbb{R}$.

Proof. By Axiom 3, $|Z(t)| \leq 2 \sum_{n=1}^{N(t)} n^{-1/2} + |R(t)| = O(t^{1/4}) + O(t^{-1/4}) = O(t^{1/4})$. By Lemma 10, $\Theta'(t) \geq c - c_* > 0$, so $|Y(u)| = |Z(\Theta^{-1}(u))|/\sqrt{\Theta'(\Theta^{-1}(u))} = O(|\Theta^{-1}(u)|^{1/4}) = O(|u|^{1/4})$ since $\Theta^{-1}(u) = O(u/\log u)$. Tempered: polynomial-growth locally-integrable functions are tempered distributions of polynomial order. $\square$

Corollary 13 (Entire extension). Y agrees a.e. on $\mathbb{R}$ with the restriction of a unique entire function $\tilde{Y} : \mathbb{C} \rightarrow \mathbb{C}$ of exponential type ≤ 1 , real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with $\text{supp } \hat{\tilde{Y}} \subseteq [-1, 1]$.

Proof. Axiom 5 gives $\text{supp } \hat{Y} \subseteq [-1, 1]$ for Y viewed as a tempered distribution (Lemma 12). Axiom 4 (Schwartz–Paley–Wiener) then produces a unique entire function $\tilde{Y}$ of exponential type ≤ 1 , of polynomial growth on $\mathbb{R}$, whose restriction to $\mathbb{R}$ agrees a.e. with Y . Reality: Z real on $\mathbb{R}$ (Axiom 1), $\sqrt{\Theta'} > 0$ (Lemma 10), so Y is real on $\mathbb{R}$; reality of the restriction forces $\tilde{Y}(\bar{z}) = \overline{\tilde{Y}(z)}$, so $\tilde{Y}$ is real on $\mathbb{R}$. $\square$

4 Laguerre–Pólya Membership

Theorem 14 ($\tilde{Y} \in \mathcal{LP}$). $\tilde{Y}$ belongs to the Laguerre–Pólya class:

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \left(1 - \frac{z}{x_n}\right) e^{z/x_n}$$

with $C \in \mathbb{R}$, $m \in \mathbb{N}_0$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$.

Proof. Step 1 (order and type). By Corollary 13, $\tilde{Y}$ is entire of exponential type ≤ 1 . Hence order $\rho(\tilde{Y}) \leq 1$, and if $\rho = 1$, finite type.

Step 2 (zero-counting bound). By Jensen's formula and exponential type ≤ 1 ,

$$\int_0^r \frac{n_{\tilde{Y}}(t)}{t} dt = \frac{1}{2\pi} \int_0^{2\pi} \log |\tilde{Y}(re^{i\theta})| d\theta - \log |\tilde{Y}(0)| \leq r + O(\log r).$$

Hence $n_{\tilde{Y}}(r) = O(r)$, and by partial summation, $\sum |z_n|^{-2} < \infty$.

Step 3 (Hadamard factorization). Since $\rho(\tilde{Y}) \leq 1$ and $\sum |z_n|^{-2} < \infty$, the Hadamard genus is ≤ 1 and

$$\tilde{Y}(z) = Cz^m e^{P(z)} \prod_n \left(1 - \frac{z}{z_n}\right) e^{z/z_n}$$

with P a polynomial of degree ≤ 1 , say $P(z) = az + b$.

Step 4 (reality). $\tilde{Y}$ real on $\mathbb{R}$ forces $\overline{\tilde{Y}(\bar{z})} = \tilde{Y}(z)$. Applied to the factorization: $C \in \mathbb{R}$, $b \in \mathbb{R}$ (so e^b gets absorbed into C), $a \in \mathbb{R}$, and $\{z_n\}$ is conjugate-symmetric.

Step 5 (reality of zeros). Corollary 13 supplies exponential type ≤ 1 , real on $\mathbb{R}$, Fourier support in $[-1, 1]$, and polynomial growth on $\mathbb{R}$ (Lemma 12). Sharp endpoint is Axiom 7. Apply Axiom 6 to $\tilde{Y}$: every zero $z_n \in \mathbb{R}$. Set $x_n := z_n \in \mathbb{R} \setminus \{0\}$.

Step 6 (no Gaussian factor). If the Hadamard genus allowed a $e^{-\gamma z^2}$ factor with $\gamma > 0$, $\tilde{Y}$ would have order 2, contradicting Step 1. So $\gamma = 0$.

Step 7 (assembly). Combining:

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \left(1 - \frac{z}{x_n}\right) e^{z/x_n},$$

$C \in \mathbb{R}$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$. This is the canonical Laguerre–Pólya form with $\gamma = 0$. Hence $\tilde{Y} \in \mathcal{LP}$. $\square$

Corollary 15 (Real zeros of $\tilde{Y}$). *Every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Laguerre–Pólya functions have only real zeros by definition (Step 5 of Theorem 14). $\square$ $\square$

5 Transfer to ζ

Theorem 16 (Zero-set transfer). $\{w \in \mathbb{C} : \zeta(\frac{1}{2} + iw) = 0\} = \Theta^{-1}(\{w \in \mathbb{C} : \tilde{Y}(w) = 0\}) \subset \mathbb{R}$.

Proof. On $\mathbb{R}$, Axiom 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$; Lemma 10 gives $Z(t) = \sqrt{\Theta'(t)} \tilde{Y}(\Theta(t))$. So on $\mathbb{R}$,

$$\zeta(\frac{1}{2} + iw) = e^{-i\vartheta(w)} \sqrt{\Theta'(w)} \tilde{Y}(\Theta(w)).$$

Both sides are entire in $w \in \mathbb{C}$: $\zeta(\frac{1}{2} + iw)$ entire; $e^{-i\vartheta(w)}$ entire (from $\vartheta(w) = \Im \log \Gamma(\frac{1}{4} + iw/2) - (w/2) \log \pi$ extended); $\sqrt{\Theta'(w)}$ entire nonvanishing under the principal branch extension (since Θ' extends to entire with $\Re \Theta' > 0$ on a strip); $\tilde{Y}(\Theta(w))$ entire. By the identity theorem, the equality extends to $\mathbb{C}$.

The exponential factor $e^{-i\vartheta(w)}$ is nonvanishing. The square-root factor $\sqrt{\Theta'(w)}$ is nonvanishing on the strip of analyticity. Hence

$$\zeta(\frac{1}{2} + iw) = 0 \iff \tilde{Y}(\Theta(w)) = 0.$$

By Corollary 15, $\tilde{Y}(\Theta(w)) = 0 \Rightarrow \Theta(w) \in \mathbb{R} \Rightarrow w \in \mathbb{R}$ (since $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ bijection and Θ extends analytically with $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$ on the strip). $\square$

6 The Riemann Hypothesis

Theorem 17 (Riemann Hypothesis). *Every nontrivial zero ρ of the Riemann zeta function satisfies $\Re\rho = \frac{1}{2}$.*

Proof. Let $\rho \in \mathbb{C}$ be a nontrivial zero of ζ : $\zeta(\rho) = 0$, $0 < \Re\rho < 1$. Write $\rho = \frac{1}{2} + iw$ with $w \in \mathbb{C}$. Then $\zeta(\frac{1}{2} + iw) = 0$, so by Theorem 16, $w \in \mathbb{R}$, i.e., $\Im(w) = 0$, i.e., $\Re\rho = \frac{1}{2}$. $\square$

References

- [1] S. Crowley, *Band-limitedness of the warped pullback Y* , Corollary 18, this repository (docs/proof_work/BandLimitedness.tex).